

Chapter 1 — Dataset formats

Match storage formats to structure, volume, and tools

Learning objectives

- Name common formats for tabular, hierarchical, and geospatial data
- Explain when CSV, SQL, HDF5, or GeoJSON is a fit
- Link format choice to downstream tooling

Why format matters

- Format is how structure is serialized on disk
- Wrong format creates friction in the pipeline
- Match format to structure, volume, and tools
- Complements the type taxonomy from the previous clip

Common formats at a glance

- **CSV** — simple tabular exchange
- **SQL tables** — queryable structured stores
- **JSON / GeoJSON** — nested or spatial features
- **HDF5** — large scientific / array datasets

Example 1.8 — SQL format

```
$ cd modules/chapter1/example8/ && bash run.sh
```

Example 1.9 — HDF5

```
$ cd modules/chapter1/example9/ && bash run.sh
```

Choosing a format

- Start from structure (tabular vs nested vs spatial)
- Consider size, update rate, and who will query it
- Prefer formats your stack already supports well
- Document the choice in metadata (next quality topics)

Takeaways

- Format = serialization of structure
- CSV and SQL dominate everyday tables
- HDF5 and GeoJSON serve specialized shapes and scales